Statistical Inference and Two-Phase Audit Design for Contaminated Measurements in Clinical Research: The catmech R Package

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Abstract

Continuous measurements in multicenter clinical trials, electronic health record (EHR) disease registries, and observational epidemiological cohorts are frequently compromised by transcription mistakes, unit conversion errors, assay calibration shifts, and missing values. Traditional statistical packages address missing covariates under Missing at Random (MAR) assumptions, discarding untrusted observations and imputing from scratch. In source-data verification (SDV) and clinical audit designs, however, recorded values are present but unreliable, and a gold-standard reference measurement is audited on only a fraction of records. Naive analysis of recorded values suffers attenuation bias, complete-case analysis of audited records inflates standard errors, and standard multiple imputation needlessly discards the precision of valid recordings. This paper presents `catmech`, an R package providing a unified, auditable framework for clinical measurement verification, two-phase prospective audit power planning, predictive calibration checking, downstream multi-endpoint regression pooling, and regulatory audit trail generation. We derive the analytical Gibbs sampling posteriors for reference-calibrated contamination mixture models, formalize prospective power calculation using two-phase influence function expansions, and establish cluster-robust sandwich covariance estimators with Barnard-Rubin finite-sample degrees of freedom for linear models (CV3 jackknife), generalized linear models (Huber-White-Liang-Zeger), and Cox proportional hazards survival models (Lin-Wei with event-based complete degrees of freedom). We evaluate the package across frozen synthetic Monte Carlo benchmarks (52,950 requests in the contamination study and over 500,000 requests across all 10 validation archives) and two large-scale real-world empirical audits: ClinicalTrials.gov ($N = 27,208$) and NHANES 2017–2018 ($N = 3,908$), alongside facility-level recounts from the Rwanda Health Management Information System (148,500 requests). Our empirical benchmarks demonstrate that retaining recorded measurements reduces root mean squared error by 43% to 48% over discarding un-audited values. We articulate an evidence-based decision rule: the contamination mixture model avoids severe attenuation bias and narrows confidence intervals by 44% relative to complete-case analysis under gross, separable recording errors, whereas simple linear imputation is preferred under subtle, diffuse discrepancies. Finally, we implement cryptographic SHA-256 provenance tracking meeting FDA 21 CFR Part 11 and ICH GCP E9 regulatory compliance standards.

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1 Introduction

In multicenter randomized clinical trials, real-world electronic health record (EHR) registries, and epidemiological cohorts, continuous biological and physiological measurements are central to primary and secondary study endpoints. Circulating biomarkers (e.g., cardiac troponin, serum creatinine, viral load), hemodynamic measurements (e.g., systolic blood pressure), and radiographic dimensions (e.g., tumor response diameters) guide clinical decisions and estimate treatment effects. However, continuous recordings in multi-site clinical registries are inevitably subject to data contamination, including assay lot calibration shifts, clerical transcription errors, decimal slips, and inconsistent unit conversions [Carroll et al., 2006, Angelopoulos et al., 2023].

To safeguard data integrity, clinical trial sponsors and regulatory authorities mandate source-data verification (SDV) and targeted clinical audits, wherein trial monitors review primary source documents (e.g., clinical chart re-abstractions, central laboratory re-assays) for a subsample of patients [U.S. Food and Drug Administration, 2003, International Conference on Harmonisation, 1998]. When discrepancies between the originally recorded measurement and the audited reference standard are uncovered, biostatisticians and trialists confront an inferential dilemma:

1. **Naively trusting recorded measurements**: Treating recorded numbers as exact introduces classic errors-in-variables. For continuous exposures, non-differential measurement error attenuates regression coefficients toward the null, obscuring true treatment effects and biological gradients [Carroll et al., 2006]. For non-linear models (e.g., logistic regression or Cox proportional hazards), measurement contamination causes complex residual confounding and severe coverage failure.
2. **Restricting analysis to audited cases (complete-case analysis)**: Analyzing only the audited subset ($\mathcal{A} \subset \{1, \ldots, N\}$, typically 10% to 25% of the cohort) protects against contamination bias but drastically reduces effective sample size, inflating standard errors, widening confidence intervals, and sacrificing statistical power [Neyman, 1938, Breslow et al., 2003].
3. **Discarding recorded values and applying standard multiple imputation**: Imputing un-audited values under standard Missing at Random (MAR) assumptions using packages such as `mice` [van Buuren and Groothuis-Oudshoorn, 2011, White et al., 2011] or `Amelia` [Honaker et al., 2011] avoids bias under correct specification. However, treating recorded values as completely unobserved discards the valuable information present in records that were recorded accurately [Rubin, 1996].

To overcome these limitations, we introduce `catmech`, an R package for auditable measurement verification, two-phase clinical audit planning, predictive calibration diagnostics, and multi-endpoint model pooling. The package unites two-phase sampling theory [Cochran, 1977, Breslow et al., 2003] with reference-calibrated Bayesian mixture modeling [Huang and Wand, 2013], Barnard-Rubin finite-sample inference [Barnard and Rubin, 1999], and cryptographic data provenance [U.S. Food and Drug Administration, 2003].

2 Statistical theory and methodology

2.1 The data generating and contamination framework

Let $i \in \{1, \ldots, N\}$ index patients enrolled across C clinical sites. Let $T_i \in \mathbb{R}$ denote the latent true biological measurement. Conditional on clinical covariates $X_i \in \mathbb{R}^p$, the latent measurement follows:

$$T_i \mid X_i \sim \mathcal{N}(X_i\beta, \sigma_T^2)$$

Let $\delta_i \in \{0, 1\}$ represent an unobserved recording contamination indicator, where $\delta_i = 1$ indicates that the recorded value was corrupted during observation or transcription. The contamination probability $\pi_i = \mathbb{P}(\delta_i = 1 \mid S_i)$ may depend on facility strata or clinical monitoring quality flags S_i. The observed recorded measurement W_i is generated as:

$$W_i = (1 - \delta_i)T_i + \delta_i(T_i + \epsilon_i), \quad \epsilon_i \sim \mathcal{N}(\mu_e, \sigma_e^2)$$

where ϵ_i represents an additive error mechanism with mean recording bias μ_e and error variance σ_e^2.

In Phase II of the audit design, an independent gold-standard reference measurement R_i is audited on a subset $\mathcal{A} \subset \{1, \ldots, N\}$ with sample size $n_{\text{audit}} = |\mathcal{A}|$:

$$R_i = T_i + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma_{\text{ref}}^2)$$

where $\sigma_{\text{ref}}^2 = 0$ corresponds to an exact, error-free recount or reference assay.

2.2 Analytical Gibbs sampling posteriors

The joint parameter vector is $\Theta = (\beta, \sigma_T^2, \pi, \mu_e, \sigma_e^2, \sigma_{\text{ref}}^2)$. Prior distributions are assigned conjugate structures: Beta priors $\pi \sim \text{Beta}(a_\pi, b_\pi)$, normal priors on coefficients $\beta \sim \mathcal{N}(\mu_\beta, \Sigma_\beta)$, and scaled inverse-$\chi^2$ priors on variance components.

For un-audited patients ($i \notin \mathcal{A}$), the conditional posterior probability of recording corruption $p_i = \mathbb{P}(\delta_i = 1 \mid W_i, X_i, \Theta)$ is evaluated analytically at each Gibbs iteration:

$$p_i = \frac{\pi f_\epsilon(W_i - \hat{T}_i)}{\pi f_\epsilon(W_i - \hat{T}_i) + (1 - \pi) f_T(W_i \mid X_i)}$$

where f_ϵ is the density of $\mathcal{N}(\mu_e, \sigma_e^2)$ and f_T is the conditional predictive density of $T_i \mid X_i$.

Conditional on $\delta_i = 0$, $T_i = W_i$ deterministically. Conditional on $\delta_i = 1$, T_i is sampled from the Gaussian posterior:

$$T_i \mid \delta_i = 1, W_i, X_i, \Theta \sim \mathcal{N}(\mu_{T|W}, \sigma_{T|W}^2)$$

$$\mu_{T|W} = \sigma_{T|W}^2 \left(\frac{X_i\beta}{\sigma_T^2} + \frac{W_i - \mu_e}{\sigma_e^2} \right), \quad \sigma_{T|W}^2 = \left(\frac{1}{\sigma_T^2} + \frac{1}{\sigma_e^2} \right)^{-1}$$

For audited patients ($i \in \mathcal{A}$), R_i directly informs T_i, yielding an exact, reference-anchored posterior. Multiple imputations $T^{(1)}, \ldots, T^{(m)}$ are drawn from post-burn-in iterations of the Markov chain.

2.3 Two-phase audit sample size and power planning

Prior to re-abstracting medical charts or commissioning secondary lab assays, investigators must budget the audit sample size n_{audit}. Using semiparametric influence function theory for two-phase estimators [Neyman, 1938, Cochran, 1977, Breslow et al., 2003, Angelopoulos et al., 2023], the asymptotic variance of an efficient parameter estimator at audit sampling fraction $f = n_{\text{audit}}/N$ satisfies:

$$\text{Var}(\hat{\theta}_f) = \frac{1}{N}\text{Var}(\psi_i) + \frac{1 - f}{fN}\text{Var}(\psi_i - \mathbb{E}[\psi_i \mid X_i])$$

where ψ_i is the efficient influence function of θ.

The precision ratio $\rho(f)$, defined as the ratio of confidence interval width at audit fraction f relative to an ideal census audit ($f = 1$), is:

$$\rho(f) = \frac{\text{Width}(f)}{\text{Width}(1)} = \sqrt{1 + k\frac{1-f}{f}}, \quad k = \frac{\text{Var}(\psi_i - \mathbb{E}[\psi_i \mid X_i])}{\text{Var}(\psi_i)}$$

Because the noise parameter k is estimated from pilot data, the realized precision ratio $\hat{\rho}(f)$ is stochastic. `catmech_audit_power()` evaluates the statistical power to achieve a target precision ratio $\rho \leq \rho_{\text{target}}$:

$$\text{Power}(n_{\text{audit}}) = \mathbb{P}\left(\sqrt{1 + \hat{k}\frac{1-f}{f}} \leq \rho_{\text{target}}\right)$$

via both closed-form non-central approximations and Monte Carlo resampling from pilot audit plans (`catmech_audit_plan()`).

2.4 Multi-endpoint model pooling with Barnard-Rubin degrees of freedom

Parameter estimates $\hat{\theta}_j$ and variance-covariance matrices $\hat{U}_j$ across $j \in \{1, \ldots, m\}$ completed datasets are combined using Rubin's rules [Rubin, 1987]:

$$\bar{\theta} = \frac{1}{m}\sum_{j=1}^{m}\hat{\theta}_j, \quad \bar{U} = \frac{1}{m}\sum_{j=1}^{m}\hat{U}_j, \quad B = \frac{1}{m-1}\sum_{j=1}^{m}(\hat{\theta}_j - \bar{\theta})(\hat{\theta}_j - \bar{\theta})^\top$$

$$T = \bar{U} + \left(1 + \frac{1}{m}\right)B, \quad \hat{\lambda} = \frac{(1 + 1/m)\text{diag}(B)}{\text{diag}(T)}$$

Standard Rubin degrees of freedom $\nu_m = (m-1)\hat{\lambda}^{-2}$ assume infinite complete-data sample size. In clinical audits where sample sizes or event counts are constrained, this assumption leads to anti-conservative confidence intervals. `catmech` implements the Barnard & Rubin (1999) small-sample adjustment [Barnard and Rubin, 1999]:

$$\nu_{obs} = \left(\frac{\nu_{com} + 1}{\nu_{com} + 3}\right)\nu_{com}(1 - \hat{\lambda}), \quad \nu_{BR} = \frac{\nu_m \nu_{obs}}{\nu_m + \nu_{obs}}$$

where ν_{com} is the complete-data degrees of freedom.

Downstream pooling covers three core clinical endpoint families:

1. **Continuous endpoints (`catmech_pool_lm`)**: Fits linear models with default delete-one-cluster jackknife (CV3) variance estimation, where $\nu_{com} = N - p$.
2. **Binary and count endpoints (`catmech_pool_glm`)**: Fits generalized linear models (logistic and Poisson) with Huber–White–Liang–Zeger cluster-robust sandwich covariance [Liang and Zeger, 1986]:

$$\hat{V}_{\text{sandwich}} = \hat{J}^{-1}\left(\sum_{c=1}^{C} s_c s_c^\top\right)\hat{J}^{-1}$$

   where $s_c = \sum_{i \in \text{cluster } c} \frac{\partial \ell_i}{\partial \theta}$ are cluster score residuals and $\hat{J}$ is the observed Fisher information matrix. Point estimates and intervals are optionally exponentiated to yield odds ratios or incidence rate ratios.
3. **Time-to-event survival endpoints (`catmech_pool_coxph`)**: Fits Cox proportional hazards models with Lin & Wei (1989) score-residual sandwich covariance [Lin and Wei, 1989]. The complete degrees of freedom are defined strictly as $\nu_{com} = d - p$, where d is the number of observed failure events and p is the parameter count. Hazard ratios $\text{HR} = \exp(\bar{\theta})$ and confidence intervals are computed using the Barnard-Rubin ν_{BR} t-distribution.

2.5 Out-of-sample predictive calibration checking

To verify that the contamination model is predictively calibrated, `catmech_holdout_check()` partitions audited records into K cross-validation folds. For each held-out record $i \in \mathcal{A}_k$, the randomized Probability Integral Transform (PIT) is computed [Dawid, 1984]:

$$u_i = F(R_i \mid \mathcal{D}_{-k}) \sim \text{Uniform}(0, 1)$$

where $F(\cdot \mid \mathcal{D}_{-k})$ is the predictive cumulative distribution function estimated from the remaining $K - 1$ folds. If the model is accurately calibrated, the PIT values follow a uniform distribution on $[0, 1]$, tested formally via a two-sided Kolmogorov–Smirnov test.

3 Software architecture and design

The `catmech` package is structured into modular S3 components designed for computational efficiency, numerical stability, and strict regulatory auditability:

Module	Core functions	Mathematical and computational role
Audit planning	`catmech_audit_plan()`, `catmech_audit_power()`	Prospective sample size determination and power curves using two-phase influence function theory.
Data diagnostics	`catmech_recording_audit()`, `catmech_holdout_check()`	Stratum-specific discrepancy auditing (Jeffreys beta intervals) and out-of-sample PIT calibration checks.
Imputation engines	`catmech_contamination()`, `catmech_reference()`	Reference-calibrated Gibbs contamination model and multilevel/marginal Gaussian imputation (`RcppArmadillo`).
Model pooling	`catmech_pool_lm()`, `catmech_pool_glm()`, `catmech_pool_coxph()`, `catmech_compare_strategies()`	Multi-endpoint pooling with Rubin rules, Barnard-Rubin df, cluster-robust sandwich covariance, and strategy comparison.
Provenance	`catmech_sensitivity()`, `catmech_audit_trail()`	Multi-scenario sensitivity envelopes and cryptographic SHA-256 regulatory compliance reporting.

A central operational requirement of regulatory clinical trials and modern observational research is that multiple imputation cannot remain restricted to continuous linear regression. In confirmatory trials, the primary estimands are predominantly time-to-event endpoints (e.g., overall survival, progression-free survival, major adverse cardiovascular events analyzed via Cox proportional hazards) or binary categorical endpoints (e.g., objective clinical response, 30-day mortality, toxicities analyzed via logistic regression). `catmech` directly integrates audit-adjusted imputation with these core clinical endpoints through `catmech_pool_coxph()` and `catmech_pool_glm()`:

- **Survival modeling (`catmech_pool_coxph`)**: Computes pooled log-hazard ratios $\bar{\theta}$ and Lin & Wei (1989) score-residual sandwich covariance [Lin and Wei, 1989], defining complete-data degrees of freedom strictly as $\nu_{com} = d - p$ (where d is the observed event count and p is the parameter count). Point estimates and Barnard-Rubin t-intervals are exponentiated to yield pooled hazard ratios (HR).
- **Binary and generalized linear modeling (`catmech_pool_glm`)**: Employs Huber–White–Liang–Zeger cluster-robust sandwich covariance [Liang and Zeger, 1986], providing robust inference against variance misspecification with exponentiated odds ratios (OR) or incidence rate ratios (IRR).
- **Unified model exploration**: All pooling functions integrate with `broom` methods (`tidy()`, `glance()`) and S3 forest-plot visualization (`plot()`), allowing clinical study teams to compare naive, complete-case, standard imputation, and mixture strategies across multiple endpoint families simultaneously.

4 Synthetic simulation studies and benchmarks

To evaluate the statistical properties of `catmech`, we conducted pre-specified Monte Carlo simulation studies across 10 validation archives (over 500,000 total requests). The primary reference-calibrated contamination study (`validation/contamination-study`) executed 52,950 planned requests (52,900 returned intervals, 16 expected design refusals, 33 MCMC convergence refusals, and 1 singular raking fit) across 4 primary cells (1,000 cohorts each) and 23 stress cells (150 cohorts each), with cohort size $N = 600$, true target slope $\beta = 0.800$, a 25% reference audit ($n_{\text{audit}} = 150$), and $m = 20$ imputations.

4.1 Scenario 1: Unflagged and gross recording contamination

We compare seven analytical strategies in two representative synthetic regimes: the primary unflagged contamination scenario (`missed_flags`, 1,000 cohorts, where recording errors occur in unflagged records) and the mixed gross-and-small error stress scenario (`error_gross_and_small`, 150 cohorts, combining gross transcription outliers with mild recording noise):

Scenario & Strategy	$\hat{\beta}$	Bias (MCSE)	Emp SD	Mean SE	Cov.	Width	RMSE
Unflagged Errors ($N_{\text{sim}} = 1,000$)							
Oracle (Latent T Known)	0.7978	-0.0022 (0.0012)	0.0394	0.0411	95.7%	0.1614	0.0394
Flag Masking (Trust Unflagged)	0.7185	-0.0815 (0.0015)	0.0472	0.0444	55.9%	0.1750	0.0942
Complete-Case (25% Audited)	0.7988	-0.0012 (0.0026)	0.0815	0.0837	95.7%	0.3309	0.0814
Discard Recorded MI	0.7896	-0.0104 (0.0021)	0.0676	0.0684	95.4%	0.2769	0.0684
Simple Linear MI	0.7806	-0.0194 (0.0018)	0.0557	0.0528	92.4%	0.2101	0.0590
Generalized Raking	0.7853	-0.0147 (0.0020)	0.0631	0.0589	92.9%	0.2331	0.0648
Contamination Mixture	**0.7952**	**-0.0048 (0.0014)**	**0.0446**	**0.0467**	**94.9%**	**0.1842**	**0.0449**
Gross & Small ($N_{\text{sim}} = 150$)							
Oracle (Latent T Known)	0.7970	-0.0030 (0.0034)	0.0411	0.0412	94.7%	0.1617	0.0411
Flag Masking (Trust Unflagged)	0.0609	-0.7391 (0.0055)	0.0674	0.0155	0.7%	0.0614	0.7422
Simple Linear MI	0.5646	-0.2354 (0.0261)	0.3200	0.0617	64.7%	0.2562	0.3964
Generalized Raking	0.5156	-0.2844 (0.0281)	0.3436	0.0693	53.7%	0.2740	0.4451
Complete-Case (25% Audited)	0.7962	-0.0038 (0.0064)	0.0787	0.0845	95.3%	0.3339	0.0785
Discard Recorded MI	0.7890	-0.0110 (0.0055)	0.0671	0.0695	95.3%	0.2816	0.0678
Contamination Mixture	**0.7883**	**-0.0117 (0.0035)**	**0.0432**	**0.0436**	**94.0%**	**0.1714**	**0.0447**

In the `missed_flags` scenario, trusting unflagged recordings (`v02_mask_normal`) suffers severe attenuation bias (-0.0815), collapsing 95% coverage to 55.9%. Complete-case analysis (`gold_only`) and discarding recorded values (`reference_normal`) remain unbiased but forfeit precision (mean CI widths of 0.3309 and 0.2769). The `catmech` contamination mixture model recovers unbiased estimates (bias: -0.0048), achieves 94.9% coverage, and narrows confidence intervals by 33.5% relative to discarding recorded values (`reference_normal`) and by 44.3% relative to complete-case analysis (`gold_only`). When gross transcription errors are mixed with small discrepancies (`error_gross_and_small`), both simple linear MI (bias: -0.2354, coverage: 64.7%) and generalized raking (bias: -0.2844, coverage: 53.7%) break down because gross outliers

corrupt the linear calibration slope, whereas `catmech_contamination()` preserves 94.0% coverage and low RMSE (0.0447).

Across the four primary 1,000-cohort cells (`clean`, `flagged_errors`, `missed_flags`, and `site_shift_declared`), `catmech_contamination()` met the pre-specified $[0.925, 0.975]$ Wilson coverage screen in three cells (96.5%, 96.3%, and 94.9% coverage) and narrowly missed the screen in `site_shift_declared` at 93.7% coverage (Wilson interval $[0.9202, 0.9505]$, bias -0.0085, MCSE 0.0016), while reducing mean interval width by 30% to 38% across all four primary cells relative to discarding recorded values.

4.2 Scenario 2: Finite-sample variance and clustered audits

In multicenter trials with C clinical sites, clustering induces correlation that can distort variance estimation. In an evaluation of 148,500 requests across clustered designs (148,486 returned intervals and 14 rank-deficiency refusals), standard cluster-robust standard errors (`cr1s`) and Satterthwaite degrees of freedom (`cr2`, which inflated interval widths by up to 95%) failed the pre-specified eligibility rules. The delete-one-cluster jackknife (`cv3`) met all eligibility criteria, restoring recount coverage to 92.0%–97.5% across 11 clinical indicators and reducing mean distance from nominal 95% coverage from 3.18 to 1.47 percentage points under random-intercept imputation, justifying its establishment as the default variance estimator in `catmech_pool_lm(cluster = )`.

5 Real-world empirical benchmarks

To test whether the advantages observed in synthetic simulations generalize to real-world data, the package was evaluated against two large-scale public datasets containing actual recording discrepancies without synthetic corruption (33,800 requests executed across 1,000 resampled cohorts of size $N = 600$ at the 25% audit fraction and 200 cohorts at secondary fractions):

Dataset & Metric	Audit 25%	Discard Rec.	Contamination	Linear MI	Full Ref. (Oracle)
ClinicalTrials.gov ($N = 27,208$)					
Enrolled vs. Started discrepancy rate: 14.2%	Bias	-0.0003	+0.0068	+0.0001	-0.0003
Target slope: 0.0357 log months / participant	Coverage	92.3%	91.2%	91.8%	91.2%
	Mean Width	0.205	0.106	0.102	0.096
	RMSE	0.0539	0.0308	0.0286	0.0275
NHANES 2017–2018 ($N = 3,908$)					
Self-reported vs. measured weight discrepancy: 67.3%	Bias	-0.0063	-0.0066	-0.0025	+0.0001
Target slope: 0.127 mmHg / kg	Coverage	94.2%	94.5%	93.9%	94.9%
	Mean Width	0.297	0.155	0.154	0.140
	RMSE	0.0785	0.0408	0.0409	0.0357

On both real datasets, retaining recorded values nearly halved the root mean squared error (RMSE reduced by 43% on ClinicalTrials.gov and 48% on NHANES) relative to discarding un-audited values. Crucially, when recording discrepancies are subtle and continuous rather than gross and bimodal, simple linear imputation (`catmech_reference(use_recorded = TRUE)`) matched or slightly outperformed the mixture model with fewer parametric assumptions.

In facility-level data verification benchmarked against the Rwanda Health Management Information System recount audit [Nshimyiryo et al., 2020] across 11 clinical indicators, `catmech`'s CV3 jackknife cluster variance achieved nominal 92.0% to 97.5% recount coverage across clinical facilities.

6 Methodological decision framework

Based on our empirical and simulation findings, we recommend the following evidence-based protocol for clinical data audits:

```
[ Clinical Measurement Discrepancy Detected ]
                        |
Is discrepancy gross, large, and separable?
 (e.g., transcription slips, lot shifts)
                 /           \
              YES             NO (rounding, diffuse noise)
              /                \
[ Use Contamination Model ]    [ Use Simple Linear MI ]
 catmech_contamination()        catmech_reference(
                                  use_recorded = TRUE)
```

1. **Gross, separable contamination**: When recording errors represent distinct corruption events (e.g., decimal transcription slips, sensor failures), the Bayesian contamination mixture model is recommended to prevent severe attenuation bias while maintaining near-oracle precision.
2. **Subtle, diffuse differences**: When errors represent continuous rounding or mild reporting bias, simple linear imputation is recommended as a parsimonious baseline.

7 Discussion: simulation-to-practice divergence and methodological implications

A principal methodological contribution of this work is illuminating the divergence between synthetic Monte Carlo simulations and empirical real-world clinical audits, an insight with direct consequences for clinical trial analysis.

In the methodological statistical literature, novel imputation models are almost universally benchmarked against synthetic datasets with clean, gross, and bimodal error structures (e.g., decimal point displacements, transcription slips, or large sensor batch shifts). Under these simulated conditions, as shown in Scenario 1 of our Monte Carlo benchmarks, the Bayesian contamination mixture model achieves substantial gains: in `error_gross_and_small`, it eliminates the severe attenuation bias that compromises simple linear imputation (bias of -0.0117 vs. -0.2354) while narrowing confidence intervals by 44.3% relative to complete-case analysis in `missed_flags` (0.1842 vs. 0.3309) and preserving nominal coverage (94.0% to 94.9%).

However, our empirical investigations across large-scale real-world registries without synthetic corruption (the ClinicalTrials.gov registry and the NHANES epidemiological audit) reveal a fundamentally different statistical landscape. Real-world recording discrepancies are rarely gross or neatly separable into discrete contaminated versus pristine sub-populations. Instead, they predominantly manifest as subtle, continuous discrepancies arising from digit preference, cognitive rounding in self-reported measures, differing clinical measurement protocols across sites, or mild procedural divergence.

In these real-world regimes, the empirical results challenge methodological dogma: simple linear imputation (`catmech_reference(use_recorded = TRUE)`) matched or slightly outperformed the mixture model in predictive accuracy (RMSE of 0.0286 vs. 0.0308 on ClinicalTrials.gov) while requiring substantially fewer parametric assumptions and zero MCMC convergence overhead.

This divergence is not a defect of the mixture engine, but a crucial empirical finding: *simulated gross errors favor mixture modeling, but real-world diffuse errors favor linear multiple imputation.* Methodologically,

this divergence demonstrates why the decision framework formalized in Section 6 is essential. Rather than assuming that a complex Bayesian mixture model is universally optimal, trialists must treat the choice of imputation model as an empirical question. `catmech` operationalizes this principle by providing:

1. Exploratory diagnostic screening via `catmech_recording_audit()` to quantify error rate and detect between-stratum heterogeneity before model fitting;
2. Out-of-sample predictive cross-validation via `catmech_holdout_check()` to guard against overfitted or miscalibrated imputation distributions;
3. Direct strategy benchmarking via `catmech_compare_strategies()` to evaluate whether data characteristics justify the mixture model's complexity or are robustly served by linear data augmentation.

By bridging prospective two-phase sample size determination, robust multi-endpoint pooling across linear, logistic, and Cox survival models, and cryptographic 21 CFR Part 11 audit trails, `catmech` equips biostatisticians with a defensible, reproducible, and mathematically grounded framework for clinical data verification.

8 Regulatory compliance and data governance

Regulatory submissions in biopharmaceutical and medical device trials (FDA 21 CFR Part 11 [U.S. Food and Drug Administration, 2003] and ICH GCP E9 [International Conference on Harmonisation, 1998]) require a tamper-evident audit trail for any algorithmic data imputation. `catmech_audit_trail()` generates complete provenance records in Markdown, JSON, or standalone HTML, capturing:

1. Full execution environment metadata (R version, operating system platform, package version, UTC timestamp).
2. Cryptographic SHA-256 checksums of the raw input dataset, model specification, and each of the m completed datasets.
3. Convergence metrics (MCMC iterations, Gelman-Rubin $\hat{R}$, bulk and tail effective sample sizes).
4. Pooled parameter estimates and multi-scenario sensitivity envelopes.

9 Quality assurance and verification

`catmech` has been developed under an automated software verification harness:

- **Unit tests**: 1,754 unit tests executed via `testthat`, achieving 0 failures, 0 warnings, and 0 skips.
- **CRAN conformance**: Fully compliant with CRAN package policies, passing `R CMD check --no-manual` with `Status: OK` (0 errors, 0 warnings, 0 notes).
- **Automated artifact verification**: Verified against an independent Python verification suite (`verify-artifacts.py`, `verify-release-archive.py`, and `verify_paper_numbers.py`) confirming exact correspondence across 10 frozen simulation and empirical benchmark archives and over 500,000 replicate requests.

10 Availability and software details

`catmech` is open-source software distributed under the MIT license. It requires R ($\geq$ 4.1.0) and a C++17 compiler, utilizing `Rcpp`, `RcppArmadillo`, `digest`, `jsonlite`, `mice`, `pan`, and `survival`. Source code, documentation, and vignettes are available at https://github.com/patrickiben/catmech.

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